Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The diagnostic combines ACPC Tail Risk (ATR), which measures high-quantile clean/noisy rollout disagreement for the same state under the same action sequence, with Selective Margin Pass Rate (SMPR), which checks that task-grounded near-boundary distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian robustness plateaus. At recovered endpoints, ATR drops sharply and SMPR rises to near one across tasks. These results support ATR and SMPR as bounded ACPC diagnostics for matched-Gaussian robustness plateaus in fixed JEPA world-model checkpoints.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose, doorway state, or object-goal relation if that distinction changes the next useful action. Encoder-level invariance is therefore an incomplete robustness target: planning uses the model's *predictions*, so the robustness question should be asked after action-conditioned rollout.

We study this gap as a controlled diagnostic problem for JEPA latent world-model control. Let a clean history and a visually perturbed history describe the same underlying state, and evaluate both branches under the same action sequence. The desired behavior is not necessarily identical encoder latents. It is that task-relevant predicted futures agree after the shared action intervention, while states that differ in action, transition, cost, or goal relation remain separable. This selective requirement is the core of action-conditioned predictive consistency (ACPC).

The selective tension matters because consistency alone can be misleading. A collapsed model can make clean and corrupted views agree while erasing distinctions needed for planning. Conversely, a large encoder shift can be harmless if the action-conditioned predictor contracts nuisance directions before they affect the predicted rollout used by planning. The diagnostic must therefore pair a same-state predictive-stability metric with an anti-collapse margin check for task-grounded near-boundary separations.

We instantiate this diagnostic with a fixed Gaussian intervention. Across PushT, TwoRoom, Reacher, and Cube, we evaluate no-noise LeWM [19] checkpoints and full-sequence input-side Gaussian-noise checkpoints over $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$. The primary endpoint is observation-only Gaussian evaluation noise with the goal image kept clean. Across three independent LeWM training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher at evaluation noise

$\sigma = 0.08$. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about fixed JEPA world-model checkpoints.

The evidence shows broad matched-Gaussian recovery plateaus rather than a stable universal point optimum. Input-side noise training recovers the observation-noise endpoint on TwoRoom, PushT, and Reacher and gives a weaker but positive Cube signal across training seeds. At recovered endpoints, ACPC Tail Risk (ATR) drops sharply and Selective Margin Pass Rate (SMPR) rises to near one across tasks. Bounded blur/resize checks delimit perturbation scope.

This paper makes three contributions. First, it formulates visual robustness for JEPA world-model control as selective action-conditioned predictive consistency rather than encoder invariance. Second, it instantiates this diagnostic with ATR for same-state predictive tail risk and SMPR for task-grounded anti-collapse margins. Third, it provides a fixed-checkpoint Gaussian robustness study across four tasks and three LeWM training seeds, with bounded unseen-stressor checks that delimit the perturbation scope.

2 Related Work

This diagnostic sits between latent-prediction representation learning and robust visual-control methods. The related work is organized around three questions: what latent prediction promises, where augmentation-based invariance is insufficient for control, and how control-relevant representation work treats downstream predictive consequences.

2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability guard. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.2 then maps these two requirements to ATR and SMPR.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{1}$$

Let Π denote a *task-relevant predictive projection* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k \, d\big(\Pi(\hat{z}_{t+k}), \, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{2}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \dots, u_H), (v_1, \dots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{3}$$

so that Equation (2) is exactly the clean/perturbed projected-rollout distance under the shared action sequence. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder geometry remains an indispensable first-stage risk signal: noisy views should stay in the same-state predictive basin and avoid crossing into task-distinct neighborhoods. The narrower point is that raw encoder distance alone is not a complete robustness criterion, because the effect of an encoder shift depends on local action-conditioned rollout sensitivity and candidate margins.

Action-relevant discriminability. Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future signal that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future signal diverges,

$$\Delta_{\mathrm{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}), \, Y^H(s_j, \mathbf{a})\big) > m, \tag{4}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i), \, \Psi(z_j)\big) > m', \tag{5}$$

where Ψ is a discriminability projection, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (2)–(5) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (5), so the two conditions are inseparable.

The same guard can be posed over state–action pairs: if two pairs have external task-relevant futures that differ by more than a margin, their projected rollouts should remain separated. This includes the same clean state evaluated under two dynamically distinct actions. The reported SMPR instantiates the guard with task-grounded near-boundary state pairs rather than adding a separate action-faithfulness metric.

3.2 Operational consistency diagnostics

The reported diagnostic uses two metrics matched to this logic. ACPC Tail Risk (ATR) summarizes the upper tail of normalized same-state clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. We use the 90th percentile in all experiments. This is a fixed empirical reporting choice, not a theoretical constant. Selective Margin Pass Rate (SMPR) is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better.

ATR tracks the high-tail term $\Pr[D > \epsilon]$ in the sampled-pool tail argument. SMPR estimates the discriminability guard by checking whether task-grounded near-boundary pairs remain separated beyond the same-state noisy radius. Thus low ATR without high SMPR is not interpreted as robustness, because collapse can also make clean and corrupted views agree. Encoder geometry and planner-side audits remain useful development checks, but they are not separate reported diagnostics here.

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{6}$$

where J is the planner cost on the projected-rollout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Encoder geometry enters the planning link through the composed rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a})).$$

Smaller same-state encoder shifts and reduced neighborhood crossing lower upstream risk, but the planning-relevant quantity is the projected-rollout distance after the shift is passed through $G_{\mathbf{a}}$. The empirical diagnostic therefore reports action-conditioned rollout-disagreement tails rather than raw encoder distances alone.

If the planner cost J is locally L_J-Lipschitz on a fixed clean/perturbed projected-rollout neighborhood, then a paired rollout discrepancy at most ϵ implies candidate-cost drift at most $L_J\epsilon$:

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \le \epsilon \quad \Rightarrow \quad |C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \le L_J\epsilon.$$

For a shared candidate set, if every candidate cost moves by at most η and the clean top-1/top-2 margin Δ is larger than 2η, the top candidate is unchanged. Equivalently, the ACPC-cost condition gives fixed-candidate stability when $\Delta > 2L_J\epsilon$. These are fixed-candidate statements only; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory. Formal statements and proofs are in Appendix A.

3.4 Sampled-pool tail motivation and Gaussian sensitivity

The fixed-candidate statements motivate the deterministic link from paired rollout agreement to candidate decisions. A once-sampled candidate pool adds a tail requirement: if a single candidate has $\Pr[D > \epsilon] \le \delta$, then a pool of size K can still contain a bad clean/noisy rollout pair with probability up to $K\delta$. A flip can also occur when the clean top-1/top-2 margin is at most $2L_J\epsilon$. This is the reason ATR uses an upper-tail summary rather than a mean disagreement score, and it is also why ATR is not a closed-loop control guarantee. Appendix A gives the sampled-pool proof.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 1 (Local Gaussian ACPC sensitivity)**.** *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi\left[\|G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2\right] = \sigma^2 \|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Noise training can improve robustness through both factors: it can reduce same-state encoder neighborhood crossing, and it can reduce the rollout sensitivity applied to any remaining encoder shift. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}} J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while projected rollouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m],$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic orientation rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

Collapse caveat. Low same-state ACPC alone is insufficient. A constant encoder–predictor can make every clean and perturbed rollout identical while also mapping action-distinct states to identical projected futures. This is why SMPR is part of the reported diagnostic: ATR measures predictive stability under nuisance perturbation, and SMPR checks that task-grounded separations have not been erased.

4 Study protocol

The protocol separates behavior endpoints from diagnostics. Closed-loop success under Gaussian evaluation noise establishes the behavioral phenomenon; ATR and SMPR are then computed on fixed checkpoints to localize whether recovered endpoints also satisfy the selective ACPC pattern. Training seed, checkpoint epoch, evaluation seeds, and trajectory counts are kept fixed across the main comparisons.

4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from Uniform$(0, \sigma_{\max})$. This is a full-sequence input-side augmentation. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean $\pm$ population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Figure 1.

The main results follow a two-part evidence chain. Closed-loop figures and tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR checks whether task-grounded near-boundary separations survive that contraction. Unseen-stressor scores are auxiliary scope checks rather than additional core diagnostics.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; checkpoint ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under observation-only Gaussian evaluation noise. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean $\pm$ population std across the three training seeds. The goal image is kept clean in every column.

Table 1: LeWM-base under observation-only Gaussian evaluation noise. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The goal image is kept clean.

Task	eval $\sigma = 0$	eval $\sigma = 0.03$	eval $\sigma = 0.05$	eval $\sigma = 0.08$
TwoRoom	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
PushT	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
Reacher	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Cube	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27

LeWM-base is strong on unperturbed images, but observation noise alone produces large closed-loop losses at $\sigma = 0.08$: PushT loses 74.56 ± 5.55 points, Reacher 41.00 ± 1.44 points, TwoRoom 25.89 ± 2.38 points, and Cube 22.11 ± 2.78 points between the first and last columns. The progression across noise levels is smooth enough to show that the endpoint is not an isolated artifact of one severity setting. PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Figure 1 shows the full LeWM Gaussian sweep aggregated across training seeds 3072/3073/3074. Each plotted point first averages evaluation seeds 42/43/44 within a training seed and then reports mean and population std across the three training seeds. The main qualitative pattern is a broad, task-dependent recovery plateau rather than a stable universal checkpoint ranking.

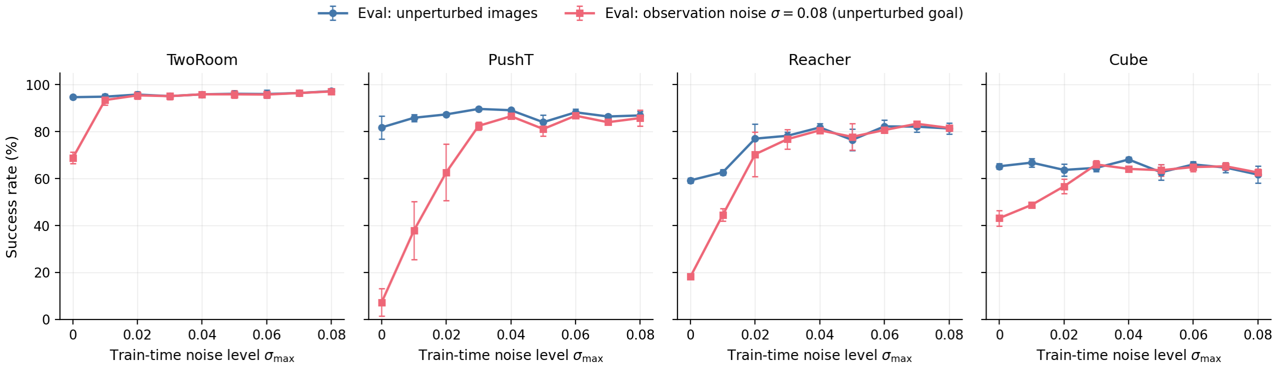

Figure 1: Three-training-seed LeWM Gaussian sweep across four tasks. Each point averages evaluation seeds 42/43/44 within each training seed and then reports mean across training seeds 3072/3073/3074; error bars show population std across training seeds. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. The curves document broad task-dependent recovery plateaus rather than checkpoint rankings.

Because Figure 1 already aggregates the full sweep across three training seeds, we do not report a separate endpoint leaderboard in the main text. The fixed $\sigma_{\max} = 0.08$ endpoint substantially improves the observation-noise endpoint on TwoRoom, PushT, and Reacher, with a weaker positive Cube signal. Exact all-seed sweep values are reported in Appendix C for reproducibility.

A three-seed blur/resize score check later in this section serves as an auxiliary severe-stressor check outside the matched-Gaussian endpoint. The next subsection asks whether the recovered Gaussian endpoints also move in the ATR/SMPR diagnostics.

5.3 ATR/SMPR selective-ACPC diagnostics

For the main diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Table 2 reports population mean/std across training seeds.

Table 2: ATR/SMPR selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

Task	ATR base	ATR std0.08	SMPR base	SMPR std0.08
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

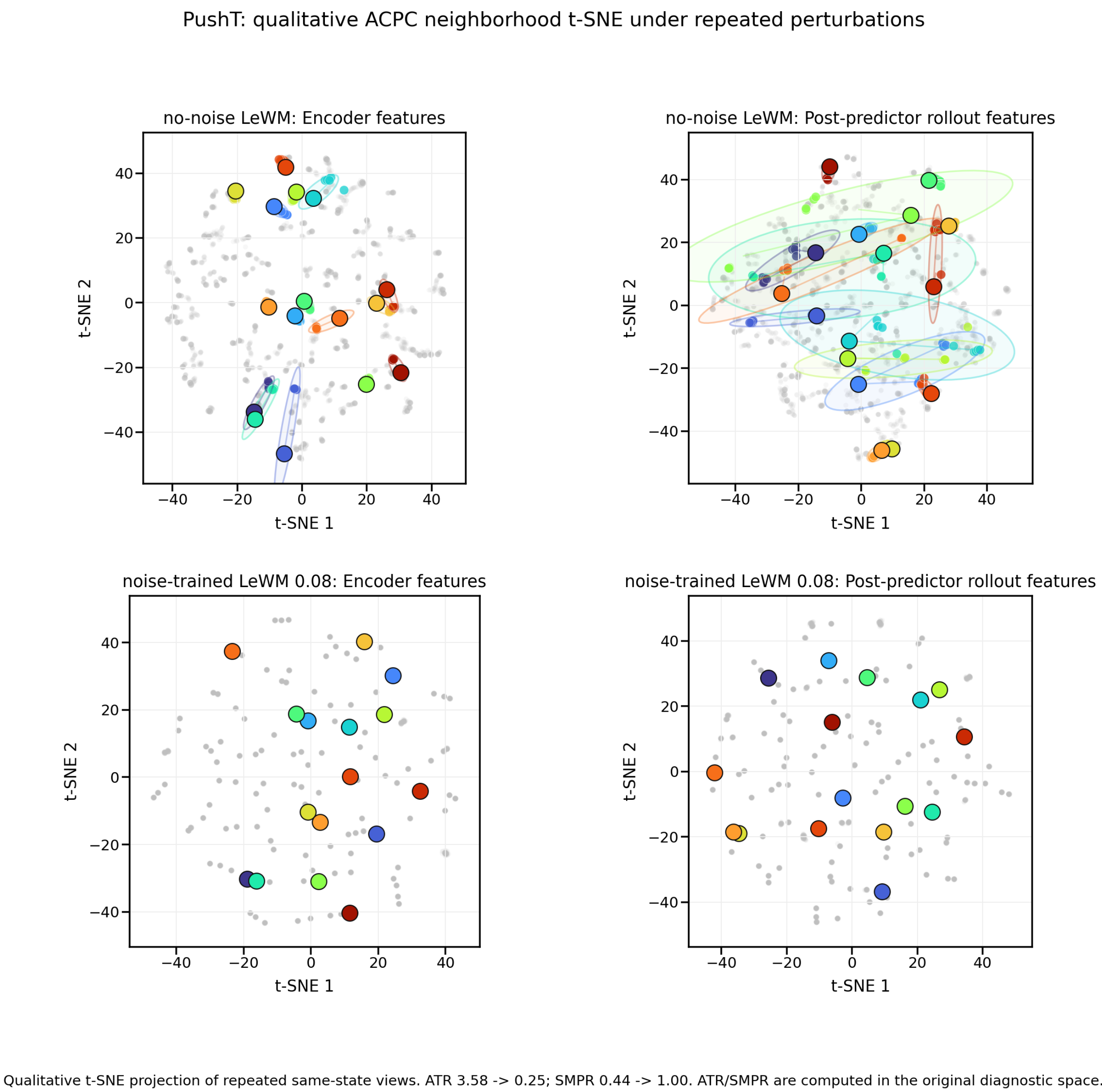

Figure 2: Qualitative PushT ACPC neighborhood t-SNE visualization for the no-noise versus $\sigma_{\max} = 0.08$ contrast. The four panels show encoder features and post-predictor rollout features for repeated same-state perturbation views; colored clusters mark fixed random anchors. The t-SNE projection is qualitative. ATR and SMPR are the quantitative diagnostics and are computed in the original projected-rollout diagnostic space.

The two diagnostics move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. Figure 2 provides qualitative visual intuition on PushT: repeated noisy views of the same state become tighter after noise training in both encoder and post-predictor feature projections. We interpret this as encoder basin repair plus further predictive contraction of residual nuisance variation, while keeping the quantitative claim on ATR, SMPR, and closed-loop behavior rather than distances in the t-SNE plane. Cube remains the useful boundary case: ATR and SMPR improve strongly, but the closed-loop Gaussian recovery in Figure 1 is weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest

35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter audit, while the high-noise endpoint reaches near-one pass rate; stronger hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation.

5.4 Bounded unseen-stressor check

The main evidence remains the matched Gaussian-noise endpoint. After freezing the Gaussian grid, we ran strongest-severity unseen-stressor score checks to test whether the fixed $\sigma_{\max} = 0.08$ endpoint transfers outside the training perturbation family. Table 3 reports behavior only. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We treat this as bounded behavior outside the matched Gaussian setting.

Table 3: Bounded unseen-stressor score check. Values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity unseen runs. The two score columns compare the no-noise LeWM checkpoint with the fixed $\sigma_{\max} = 0.08$ noise-trained checkpoint under the same non-Gaussian evaluation stressor.

Task	evaluation stressor	no-noise ckpt	$\sigma_{\max} = 0.08$ ckpt
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57

The pattern is mixed outside the training perturbation family: the $\sigma_{\max} = 0.08$ checkpoint gives large improvements for TwoRoom and Reacher under blur, while PushT and Cube are essentially unchanged under the resize stressor. These rows therefore delimit the Gaussian result rather than extending it into a general perturbation-transfer claim.

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR checks whether same-state noisy views have small action-conditioned predictive tails; SMPR checks whether task-grounded near-boundary distinctions survive that contraction. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

Scope. Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. ATR/SMPR are not a checkpoint-ranker for $\sigma_{\max}$ and do not replace closed-loop evaluation inside a broad plateau. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

What remains. The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. Stronger discriminability evidence would require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR checks task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian robustness plateaus across three training seeds, and ATR/SMPR move in the expected direction at recovered endpoints.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

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A Proofs for ACPC Diagnostics

These proofs support the diagnostic use of ATR and SMPR; they do not provide closed-loop control guarantees.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that J is locally L_J-Lipschitz on the clean/perturbed projected-rollout neighborhood evaluated by the fixed candidate set under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h),\mathbf{a})),\Pi(F_\theta^{1:H}(E_\theta(\tilde{h}),\mathbf{a}))\big) \le \epsilon,$$

then

$$|C_h(\mathbf{a},g) - C_{\tilde{h}}(\mathbf{a},g)| \le L_J\epsilon.$$

Proof. The planner cost J is L_J-Lipschitz on the projected-rollout sequence, so the cost difference is bounded by L_J times the clean/perturbed projected-rollout distance. □

Proposition 3 (Candidate top-1 stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1,\dots,\mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j,g) - C_{\tilde{h}}(\mathbf{a}^j,g)| \le \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)},g) - C_h(\mathbf{a}^{(1)},g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j,g) - C_{\tilde{h}}(\mathbf{a}^{(1)},g) \ge C_h(\mathbf{a}^j,g) - C_h(\mathbf{a}^{(1)},g) - 2\eta \ge \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Theorem 1 (Sampled-pool ACPC tail bound)**.** *Let $\mathcal{A} = \{\mathbf{a}^1,\dots,\mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool, and use deterministic tie-breaking for* $\arg\min$*. For each sampled candidate define*

$$D_j = d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h),\mathbf{a}^j)),\Pi(F_\theta^{1:H}(E_\theta(\tilde{h}),\mathbf{a}^j))\big)\,.$$

Assume J is locally L_J-Lipschitz on the evaluated projected-rollout neighborhood and that a single draw from q satisfies

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \le \delta.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)},g) - C_h(\mathbf{a}^{(1)},g)$ be the clean branch top-1/top-2 margin inside the sampled pool. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j,g) \neq \arg\min_j C_{\tilde{h}}(\mathbf{a}^j,g)\right] \le K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \le 2L_J\epsilon].$$

Proof. For a sampled pool of size K, the probability that at least one candidate has clean/noisy rollout discrepancy above ϵ is at most $K\delta$ by the union bound. On the complement of that event, every candidate cost moves by at most $L_J\epsilon$. A flip can then occur only if the clean top margin is at most $2L_J\epsilon$, which contributes the margin-tail term. This proves the stated sampled-pool instability bound. □

Local Gaussian sensitivity. For small Gaussian pixel noise ξ, a first-order expansion gives $E_\theta(o+\xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Details

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$.

ATR is computed at each task–training-seed–checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

C Additional Gaussian Evaluation Tables

These tables report the full observation-only Gaussian evaluation columns available for the three-training-seed sweep behind Figure 1. Each training-seed value first averages evaluation seeds 42/43/44; table cells report mean $\pm$ population std across training seeds 3072/3073/3074.

Table 4: Full three-training-seed LeWM observation-only Gaussian sweep evaluation table. Columns report clean evaluation and observation-only Gaussian evaluation at $\sigma = 0.03/0.05/0.08$ with a clean goal image. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value first averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed.

Task	training $\sigma_{\max}$	clean eval	obs-only $\sigma = 0.03$ eval	obs-only $\sigma = 0.05$ eval	obs-only $\sigma = 0.08$ eval
TwoRoom	0.00	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
TwoRoom	0.01	94.89 ± 1.10	95.00 ± 0.82	94.78 ± 1.77	93.44 ± 2.08
TwoRoom	0.02	95.78 ± 1.10	95.89 ± 1.29	95.22 ± 1.29	95.44 ± 1.57
TwoRoom	0.03	95.11 ± 0.87	95.00 ± 0.82	95.22 ± 0.83	95.11 ± 1.34
TwoRoom	0.04	95.89 ± 0.42	96.44 ± 0.96	95.78 ± 0.31	95.89 ± 0.57
TwoRoom	0.05	96.11 ± 1.50	96.00 ± 1.91	96.00 ± 1.52	95.89 ± 1.55
TwoRoom	0.06	96.00 ± 1.70	95.56 ± 1.81	95.56 ± 1.40	95.78 ± 1.55
TwoRoom	0.07	96.44 ± 0.63	96.33 ± 0.54	96.11 ± 0.68	96.44 ± 0.31
TwoRoom	0.08	97.22 ± 0.83	97.22 ± 0.68	97.33 ± 0.72	97.11 ± 0.57
PushT	0.00	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
PushT	0.01	85.89 ± 1.50	82.22 ± 3.03	70.33 ± 6.33	37.89 ± 12.37
PushT	0.02	87.33 ± 0.72	85.11 ± 1.13	82.89 ± 2.69	62.67 ± 11.98
PushT	0.03	89.67 ± 0.00	89.67 ± 0.27	88.78 ± 2.08	82.44 ± 1.66
PushT	0.04	89.11 ± 0.83	88.22 ± 1.13	87.33 ± 1.19	86.56 ± 0.16
PushT	0.05	84.00 ± 3.03	84.22 ± 2.53	82.56 ± 3.13	81.11 ± 3.03
PushT	0.06	88.22 ± 1.34	88.89 ± 1.03	87.33 ± 0.47	86.78 ± 0.63
PushT	0.07	86.44 ± 0.87	86.00 ± 1.19	84.22 ± 0.42	84.00 ± 0.27
PushT	0.08	86.89 ± 1.59	86.56 ± 2.20	84.67 ± 2.88	85.78 ± 3.45
Reacher	0.00	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Reacher	0.01	62.67 ± 1.19	60.67 ± 1.09	56.00 ± 0.72	44.56 ± 2.57
Reacher	0.02	77.00 ± 6.18	76.44 ± 8.17	77.22 ± 6.68	70.33 ± 9.46
Reacher	0.03	78.22 ± 1.66	79.33 ± 0.72	79.11 ± 1.10	76.78 ± 4.17
Reacher	0.04	81.78 ± 1.59	82.00 ± 1.52	81.11 ± 2.62	80.56 ± 1.03
Reacher	0.05	76.44 ± 4.57	77.33 ± 5.76	77.56 ± 4.01	77.78 ± 5.67
Reacher	0.06	82.22 ± 2.79	79.33 ± 2.42	81.67 ± 1.63	80.67 ± 1.19
Reacher	0.07	82.11 ± 2.20	82.44 ± 1.29	81.44 ± 1.85	83.33 ± 1.09
Reacher	0.08	81.33 ± 2.33	82.56 ± 2.08	82.00 ± 1.78	81.56 ± 0.87
Cube	0.00	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27
Cube	0.01	66.78 ± 1.81	67.00 ± 2.13	67.56 ± 2.27	48.78 ± 0.31
Cube	0.02	63.67 ± 2.62	61.44 ± 3.45	63.33 ± 1.96	56.67 ± 3.07
Cube	0.03	64.56 ± 1.66	63.44 ± 3.03	65.33 ± 0.94	66.00 ± 1.70
Cube	0.04	68.11 ± 1.03	65.11 ± 1.50	65.44 ± 0.57	64.11 ± 0.83
Cube	0.05	62.67 ± 3.40	63.33 ± 2.42	60.67 ± 3.09	63.56 ± 2.53
Cube	0.06	66.00 ± 1.19	65.89 ± 1.03	64.67 ± 1.09	64.89 ± 1.85
Cube	0.07	64.67 ± 2.16	64.67 ± 1.89	65.44 ± 2.57	65.22 ± 1.55
Cube	0.08	61.67 ± 3.57	61.78 ± 4.16	61.89 ± 2.45	62.56 ± 1.55